Assignment – II

Database Management System

(24CSH-298)

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Submitted To:

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Ques 1: Explain each type of security Issues.

Ans: Database security issues refers to risk or problems that may lead to unauthorized access, misuse, or loss of data. These issues can be classified into following types:

1. Legal and ethical Issues:

legal and ethical concern that laws, regulations, and moral responsibilities related to the shortage and use of data in a database system. Organizations must ensure that they comply with data protection laws and privacy regulations. sensitive information such as personal data, financial records and confidential buisness information should not be accessed and shared without proper authorization.

Example: Leaking a customer personal information or using employe data without concent

2. Policy Issues:

Policy issues related to the rules and guidelines established by an organization to protect its database systems. Organizations develop security policies that defines how data should be accessed, used and protected. These policies specify user roles, access rights, responsibility of employees when handling data.

3. System Related Issue :

System Related issues arise from technical weakness in database system or infrastructure like weak authentication systems, software vulnerabilities improper configuration of database servers and lack of updates or security patches

Example: A hacker exploiting a database vulnerability to steal information.

Ques2 : Explain the different database security control techniques.

Answer : Database security control techniques are methods used to protect database from unauthorized access and misuse.

i) Access Control

Access Control restricts who can access the database and what operations they can perform, Users are given specific permission (read, write, delete). Implemented using user accounts. Ensure that only authorized users can access sensitive data.

ii) Inference Control

Inference control prevents users from deducing sensitive information from accessible data.

iii) How Control

How Control ensures that information docsnot flow from higher security levels to lower security levels,

It restricts how information moves within a systems.

iv) Data encryption

It converts plain text data into coded from cipher text to protect it from unauthorized access

Ques 3: Consider a table employee (EmpID, EmpName, Salary, DeptID). Write an SQL Query to create a view that displays only EmpID, EmpName and DeptID from the employee table.

Ans:

SQL Query:

```
CREATE VIEW employee_View AS
SELECT EmpID, EmpName, DeptID
FROM employee
```

This view will display only EmpID, EmpName, and DeptID, hiding other columns such as Salary.